Lorenzo Sarnataro

Curriculum Vitæ

Personal information

Address: Department of Mathematics
University of Toronto
40 St George Street
Toronto, ON M5S 2E4, Canada

Email: lorenzo.sarnataro@libero.it
lorenzo.sarnataro@utoronto.ca

Appointments

2024 – present **University of Toronto**, *Brauer Postdoctoral Fellow*
Mentors: Prof. Yevgeny Liokumovich and Prof. Robert Haslhofer.

Education

2017 – 2018 and 2020 – 2024 **Ph.D in Mathematics**, *Princeton University*, USA
Advisor: Prof. Fernando Codá Marques.
Thesis: Two regularity results in the theory of minimal hypersurfaces

2013 – 2017 **Master of Mathematics (MMath)**, *University of Oxford*, UK
Distinction in Prelims (first year exams)
First class Honours in Part A and Part B (second and third year exams)
First class Honours in Part C (fourth year exams)

Publications and Preprints

2026 *A uniqueness theorem for Allen–Cahn global minimizers asymptotic to the Simons cone* (with Sanghoon Lee) – preprint, 46 pages, (arXiv:2611.13435)

2026 *Improved Index and Intersection bounds for Min-Max Geodesics* (with Jared Marx-Kuo and Douglas Stryker) – preprint, 9 pages, (arXiv:2611.08450)

2026 *On the Gage–Hamilton–Grayson Theorem for free boundary curves* (with Yueheng Bao) – preprint, 14 pages, (arXiv:2610.09557)

2026 *Regularity and compactness theory of stable free boundary minimal hypersurfaces* (with Costante Bellettini, Martin Li, and Davide Parise) – preprint, 61 pages, (arXiv:2610.09596)

2026 *Boundary behaviour of stable Allen–Cahn limit interfaces in Riemannian manifolds* (with Martin Li and Davide Parise) – preprint, 60 pages, (arXiv:2610.04833)

2026 *Second order estimates on transition layers for the Allen–Cahn equation with homogeneous Neumann boundary conditions: Part I – Boundary-orthogonal transition layers* (with Akashdeep Dey, Wenkui Du, and Davide Parise) – preprint, 107 pages (arXiv:2609.32829)

2026 *Existence of free boundary minimal disks in convex regions* (with Douglas Stryker, Zhichao Wang, and Xin Zhou) – preprint, 22 pages (arXiv:2606.02396) – submitted to **J. Eur. Math. Soc.**

2024 *Index, Intersections, and Multiplicity of Min-Max Geodesics* (with Jared Marx-Kuo and Douglas Stryker), **Advances in Mathematics**, Volume 502 (2026), Part C. Article 111193, pages 1–46, (arXiv:2410.02580)

2023 *Boundary behavior of limit-interfaces for the Allen-Cahn equation on Riemannian manifolds with Neumann boundary condition* (with Martin Li and Davide Parise), **Arch. Rational Mech. Anal.** 248, **124** (2024), pages 1–43, (arXiv:2312.07210)

2023 *Existence of closed embedded curves of constant curvature via min-max* (with Douglas Stryker), **J. Reine Angew. Math. (Crelles Journal)**, vol. 2026, no. 831, pages 185–211, (arXiv:2306.04840)

2023 *Optimal regularity for minimizers of the prescribed mean curvature functional over isotopies* (with Douglas Stryker), **Camb. J. Math. 13** (2025), no. 3, pages 609–706, (arXiv:2304.02722)

Scholarships and Awards

2021-2024 *Bradford H. Arnold *42 and Mary Ellen Arnold Fellowship* (Princeton University)

2016 *Junior Mathematical Prize* (Mathematical Institute, University of Oxford) for combined performance in Part A and Part B

2014 - 2017 *Worcester College's Scholarship* for Distinction in Prelims, Part A and Part B

2014 *Fondazione M. Gaia Scholarship* for outstanding academic performance, worth €10,000 over five years

Teaching

University of Toronto:

Winter 2027 *Instructor* for *MATA37 (Calculus II for Mathematical Sciences)* at UTSC

Fall 2025 *Instructor* for *MATA31 (Calculus I for Mathematical Sciences)* at UTSC

Winter 2024 *Instructor and course head* for *MATA31 (Calculus I for Mathematical Sciences)* at UTSC

Princeton University:

Spring 2023 *Teaching assistant* for *MAT 355: Introduction to Differential Geometry*

Fall 2023 *Teaching assistant* for *MAT 300: Multivariable Analysis I*

Spring 2023 *Precept instructor* for *MAT 175: Mathematics for Economics and Life Sciences*

Fall 2022 *Precept instructor* for *MAT 175: Mathematics for Economics and Life Sciences*

Spring 2022 *Precept instructor* for *MAT 201: Multivariable Calculus*

Fall 2021 *Precept instructor* for *MAT 201: Multivariable Calculus*

Spring 2021 *Teaching assistant* for *MAT 201: Multivariable Calculus*

Fall 2020 *Teaching assistant* for *MAT 300: Multivariable Analysis I*

Invited Talks

November 2026 *Analysis and Geometric Analysis Seminar* at *Cornell University*

September 2026 *Geometric Analysis Seminar* at *Columbia University*

June 2026 *International geometric analysis conference in Milan II* at *Bocconi University*

May 2026 *Analysis Seminar* at *ETH*

February 2026 *Geometry Seminar* at *Stanford University*

February 2026 *Differential Geometry Seminar* at *UC Berkeley*

January 2026 *Workshop on Connections among Spin Geometry, Minimal Surfaces and Relativity* at *BIRS-CMI*

December 2025 *Session on Geometric PDEs* at *Canadian Mathematical Society Winter Meeting*

October 2025 *Analysis and Gravitation Seminar* at *Queen Mary University London*

September 2025 *Geometry Seminar* at *University of Waterloo*

December 2024 *Geometric Analysis Seminar* at *University of Tennessee Knoxville*

September 2024 *Geometry and Topology Seminar* at *University of Toronto*

June 2024 *Fields Institute Geometric Analysis Colloquium*

May 2024 *Informal Geometric Analysis Seminar* at *Northwestern University*

April 2024 *Analysis and PDE Seminar* at *Notre Dame University*

March 2024 *Geometric Analysis Seminar* at *CUNY*

February 2024 *Geometry and Topology Seminar* at *University of Pennsylvania*

January 2024 *Geometry Seminar* at *University of Copenhagen*

December 2023 *Analysis Seminar* at *ETH*
December 2023 *Differential Geometry Seminar* at *UCSD*
December 2023 *Geometry and Topology Seminar* at *Caltech*
November 2023 *Geometric Analysis Seminar* at *University of Chicago*
November 2023 *Geometric Analysis Seminar* at *MIT*
November 2023 *Recent advances in geometric analysis* at *CIRM*
October 2023 *Geometric Analysis Seminar* at *Rutgers University*
October 2023 *Geometry Seminar* at *Rice University*
September 2023 *Analysis and Geometric Analysis Seminar* at *Cornell University*
April 2023 *Graduate Student Seminar* at *Princeton University*

Service

Referee for: *Journal of the European Mathematical Society, International Mathematics Research Notices, Journal of Differential Geometry, Calculus of Variations and PDEs, Transactions of the American Mathematical Society.*

References

Costante Bellettini
Department of Mathematics
University College London
c.bellettini@ucl.ac.uk

Otis Chodosh
Department of Mathematics
Stanford University
ochodosh@stanford.edu

Joaquim Serra
Department of Mathematics
ETH
joaquim.serra@math.ethz.ch

Fernando Codá Marques
Department of Mathematics
Princeton University
coda@math.princeton.edu

Yevgeny Liokumovich
Department of Mathematics
University of Toronto
ylio@math.toronto.edu

Martin Man-Chun Li
Department of Mathematics
The Chinese University of Hong Kong
martin.li@cuhk.edu.hk

Robert Haslhofer
Department of Mathematics
University of Toronto
roberth@math.toronto.edu

Michael Scott Cavers (teaching)
Department of Mathematics
University of Toronto (Scarborough Campus)
michael.cavers@utoronto.ca